

Lab 12: Fast LRC Decay

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Jaysen Tsao

Lab Partner: Alex Yue

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Professor Szwerc

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Pre-lab Questions

Derive the relationship between the time constant in an LRC circuit and the half life of the voltage across the resistor.

From the general solution for the voltage across the resistor in an LRC circuit, we have:

$$V(t) = V_0 e^{-\frac{R}{2L}t} \cos(\omega t + \varphi),$$

where $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$ is the angular frequency of the oscillation. The time constant τ is defined as the time it takes for the voltage to decay to $1/e$ of its initial value, which occurs when the exponential is $1/e$:

$$e^{-\frac{R}{2L}\tau} = \frac{1}{e} \rightarrow -\frac{R}{2L}\tau = -1 \rightarrow \tau = \frac{2L}{R}.$$

This allows us to rewrite the voltage as:

$$V(t) = V_0 e^{-t/\tau} \cos(\omega t + \varphi).$$

The half-life $t_{1/2}$ is defined as the time it takes for the voltage to decay to half of its initial value, which occurs when the exponential is $1/2$:

$$e^{-t_{1/2}/\tau} = \frac{1}{2} \rightarrow -t_{1/2}/\tau = \ln\left(\frac{1}{2}\right) \rightarrow t_{1/2} = \tau \ln 2 = \frac{2L \ln 2}{R}.$$

Part 1: Driven LRC Series Circuit

An LC circuit with an inductor of labeled inductance 22 mH and a capacitor of labeled capacitance $0.01 \mu\text{F} = 10 \text{ nF}$ is driven by a function generator with a square wave output, with some internal resistance R .

Inductor value (measured)	$L \approx 22.8 \text{ mH}$
Capacitor value (measured)	$C \approx 9.6 \text{ nF}$
Total resistance (measured)	$R \approx 36.4 \Omega + 50 \Omega = 86.4 \Omega$

From the derivation in the instructions, the critical resistance of the circuit is given by:

$$R_{\text{crit}} = 2\sqrt{\frac{L}{C}} \approx 3082.2 \Omega.$$

Our measured total resistance R is much less than the critical resistance R_{crit} , so the circuit is severely underdamped. The damped angular frequency is given by:

$$\omega_d = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \approx 67\,565 \text{ rad/s}.$$

Thus, the theoretical oscillation frequency of the circuit is given by:

$$f_d = \frac{\omega_d}{2\pi} \approx 10\,753 \text{ Hz}.$$

Using the derivation from the pre-lab, the theoretical half-life of the voltage across the resistor is:

$$t_{1/2} = \frac{2L \ln 2}{R} \approx 0.3658 \text{ ms} = 365.8 \mu\text{s}.$$

	Theoretical	Experimental	% Error
Oscillation Frequency, f	10 753 Hz	$T = (1.9 \text{ div})(50 \mu\text{s/div}) = 95 \mu\text{s}$ $f = 1/T = 10\,526 \text{ Hz}$	2.115%
Half life, $t_{1/2}$	365.8 μs	$(2 \text{ div})(100 \mu\text{s/div}) = 200 \mu\text{s}$	45.329%

Part 2: Different Combinations of L and C

Case 1: Decreased L , Same C

Inductor value (measured)	$L \approx 3.3 \text{ mH}$
Capacitor value (measured)	$C \approx 9.6 \text{ nF}$

Since $f_d \propto \omega_d \propto 1/\sqrt{L}$, decreasing L while keeping C the same should increase the oscillation frequency of the circuit. The half-life should decrease, since $t_{1/2} \propto L$.

Calculating the new theoretical oscillation frequency gives:

$$f_d = \frac{\omega_d}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \approx 28\,199 \text{ Hz.}$$

Calculating the new theoretical half-life gives:

$$t_{1/2} = \frac{2L \ln 2}{R} \approx 0.0529 \text{ ms} = 52.9 \mu\text{s}.$$

Indeed, the oscillation frequency has increased and the half-life has decreased.

	Theoretical	Experimental	% Error
Oscillation Frequency, f	28 199 Hz	$T = (1.8 \text{ div})(20 \mu\text{s/div}) = 36 \mu\text{s}$ $f = 1/T = 27\,777 \text{ Hz}$	1.496%
Half life, $t_{1/2}$	52.9 μs	$(3.4 \text{ div})(20 \mu\text{s/div}) = 68 \mu\text{s}$	28.426%

Case 2: Original L , Increased C

Inductor value (measured)	$L \approx 22.8 \text{ mH}$
Capacitor value (measured)	$C \approx 100 \text{ nF}$

Since $f_d \propto \omega_d \propto 1/\sqrt{C}$, increasing C while keeping L the same should decrease the oscillation frequency of the circuit. The half-life should not change, since $t_{1/2}$ does not depend on C .

$$f_d = \frac{\omega_d}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \approx 3319 \text{ Hz.}$$

Calculating the new theoretical half-life gives:

$$t_{1/2} = \frac{2L \ln 2}{R} \approx 0.3658 \text{ ms} = 365.8 \mu\text{s}.$$

	Theoretical	Experimental	% Error
Oscillation Frequency, f	3319 Hz	$T = (1.6 \text{ div})(200 \mu\text{s/div}) = 320 \mu\text{s}$ $f = 1/T = 3125 \text{ Hz}$	5.858%
Half life, $t_{1/2}$	365.8 μs	$(1.2 \text{ div})(200 \mu\text{s/div}) = 240 \mu\text{s}$	34.395%

Part 3: Finding Critical Resistance using Potentiometer

Inductor value (measured)	$L \approx 22.8 \text{ mH}$
Capacitor value (measured)	$C \approx 9.6 \text{ nF}$
Critical resistance (measured)	$R_{\text{crit}} \approx 2440 \, \Omega + 86.4 \, \Omega = 2526.4 \, \Omega$

With the original L and C , the theoretical critical resistance is given by:

$$R_{\text{crit}} = 2\sqrt{\frac{L}{C}} \approx 3082.2 \, \Omega.$$

This is a 18.033% error compared to the theoretical critical resistance.

When R is less than the critical resistance, the circuit is underdamped and the potential across the resistor oscillates. When R is greater than the critical resistance, the circuit is overdamped and the potential across the resistor decays to zero more slowly without oscillating.

Note: As R nears zero, the decaying effect becomes less apparent, and our circuit begins to model a pure LC circuit, which exhibits SHM.

Conclusion

1. In part 1, you were asked to select an appropriate frequency for the square wave. How do you know that you have an appropriate frequency?

To see a full “damped oscillation” on the oscilloscope, we should select a frequency where the period of the square wave is long enough to see the voltage across the resistor decay. We used a period of 5 time constants, which is around $T = 5\tau = 10L/R = 6.26$ ms. This corresponds to a frequency of around $f = 1/T = 160$ Hz.

2. In part 3, how do you know when your circuit is underdamped, critically damped, and overdamped?

When the circuit is underdamped, the potential across the resistor oscillates while decaying. When the circuit is critically damped, the potential across the resistor decays to zero as quickly as possible without oscillating. When the circuit is overdamped, the potential across the resistor decays to zero more slowly without oscillating.

3. In part 3, what happens to the potential across the resistor as the resistance in the circuit is increased from its lowest to its highest value?

At low resistance values, the potential across the resistor oscillates while decaying. As the resistance increases, the oscillations become less pronounced and the potential decays more slowly. At the critical resistance, the potential decays to zero as quickly as possible without oscillating. Beyond the critical resistance, the potential decays to zero more slowly without oscillating.

4. There are multiple frequencies that are mentioned in this lab. The driven frequency is what you set with the function generator. The oscillation frequency (or the response frequency) is what is observed in the LRC circuit. How are these frequencies related?

The driven frequency is the frequency of the input signal from the function generator, i.e. how often the voltage source oscillates between “high” and “low”. The oscillation frequency is the natural frequency of the LRC circuit, which is determined by the values of L , C , and R . These frequencies are related since we must ensure that the driven frequency is low enough to allow us to see the oscillations in the circuit. If the driven frequency is too high, we may not be able to see the oscillations clearly on the oscilloscope, since the voltage across the resistor may not have enough time to decay before the next cycle of the square wave begins.

5. Why does the value of the resistor change the oscillation of the potential? Your answer should include references to the power input to the circuit from the generator, and the power dissipated in the resistor.

The resistor dissipates power as heat (such power depends on resistance via $P = I^2 R$), which reduces the energy available for oscillation in the circuit. As the resistance increases, more power is dissipated in the resistor, which leads to a faster decay of the oscillations. At low resistance values, the power input from the generator is sufficient to sustain oscillations for a longer period of time, resulting in a slower decay.